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Lab 02 DNS

All screeshot provided in each Qustion

1. Nslookup

1. Run nslookup to obtain the IP address of a Web server in Asia. What is the IP address of that server?

Here are multiple addresses shown:

108.138.94.105
108.138.94.127
108.138.94.54
108.138.94.74

```
❯ nslookup www.nifty.com

Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:    www.nifty.com
Address: 108.138.94.105
Name:    www.nifty.com
Address: 108.138.94.127
Name:    www.nifty.com
Address: 108.138.94.54
Name:    www.nifty.com
Address: 108.138.94.74
```

2. Run nslookup to determine the authoritative DNS servers for a university in Europe.

I ran the nslookup to determine the authoritative DNS servers for Oxford University is ~~primary.dns.cam.ac.uk~~ as below.

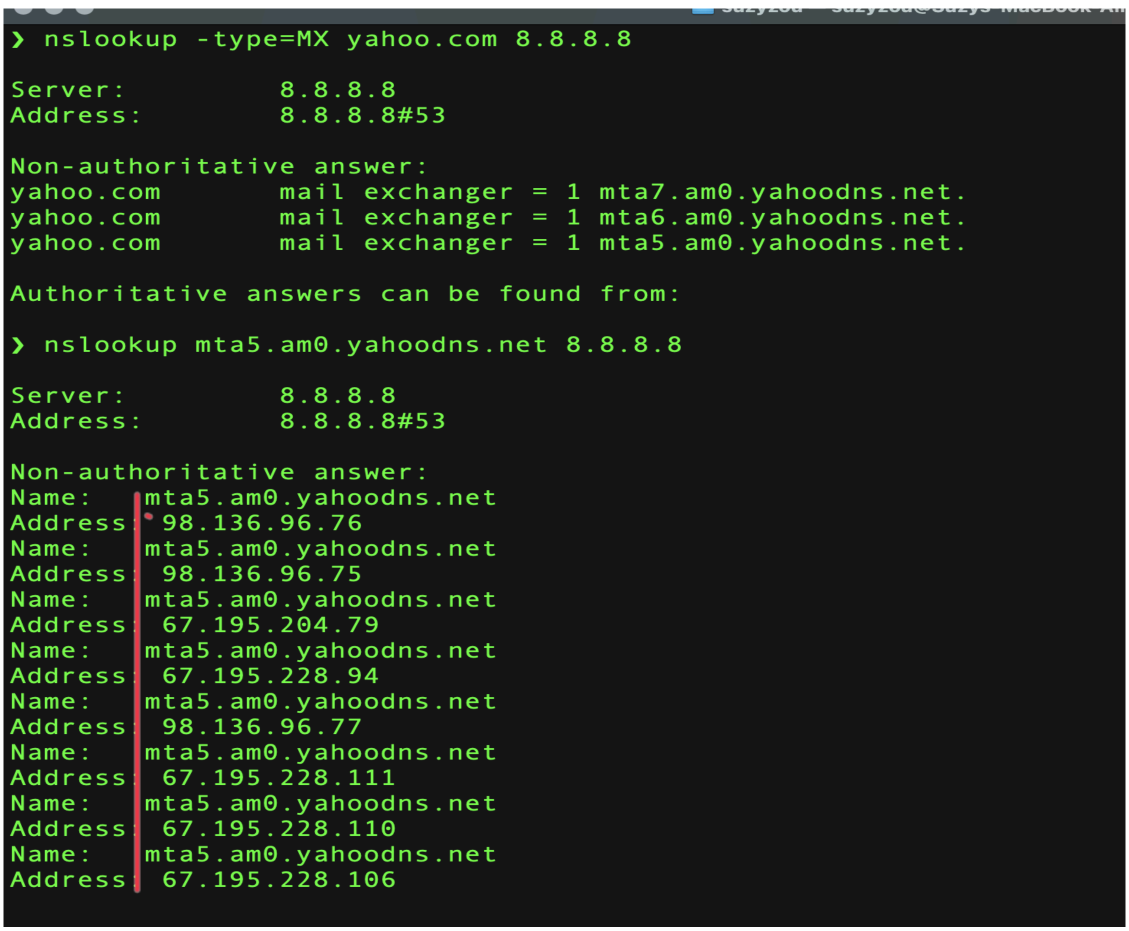
dns.ox.ac-UK

```
❯ nslookup -type=NS ox.ac.uk dns0.ox.ac.uk

Server:         dns0.ox.ac.uk
Address:        129.67.1.190#53

ox.ac.uk        nameserver = auth5.dns.ox.ac.uk.
ox.ac.uk        nameserver = auth6.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns0.ox.ac.uk.
ox.ac.uk        nameserver = dns2.ox.ac.uk.
ox.ac.uk        nameserver = auth4.dns.ox.ac.uk.
ox.ac.uk        nameserver = dns1.ox.ac.uk.
```

3. Run nslookup so that one of the DNS servers obtained in Question 2 is queried for the mail servers for Yahoo! mail. What is its IP address?

98.136.96.76
98.136.96.75
67.195.204.79
67.195.228.94
98.136.96.77
67.195.228.111
67.195.228.110
67.195.228.106

Note: Oxford's DNS refuses external queries, so I use public resolver



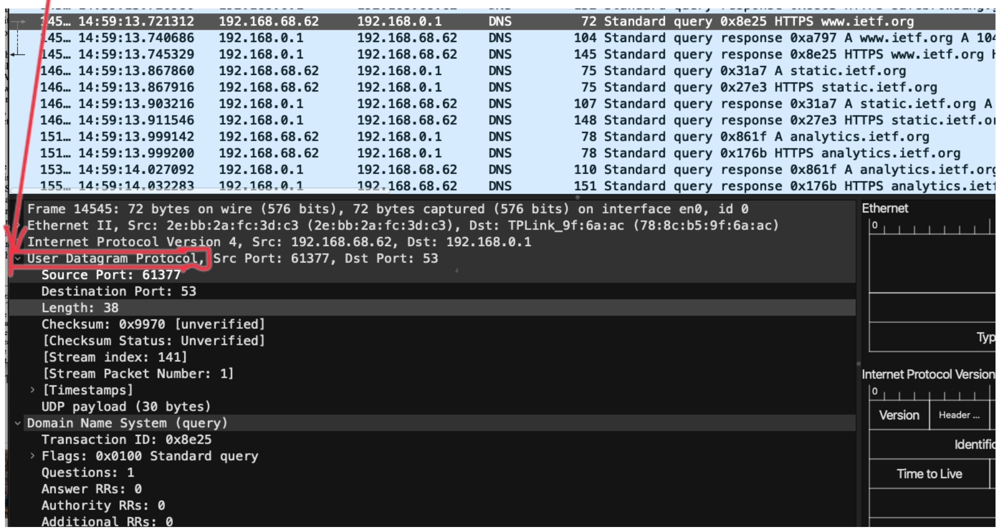
```
❯ nslookup -type=MX yahoo.com 8.8.8.8

Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
yahoo.com       mail exchanger = 1 mta7.am0.yahoodns.net.
yahoo.com       mail exchanger = 1 mta6.am0.yahoodns.net.
yahoo.com       mail exchanger = 1 mta5.am0.yahoodns.net.

Authoritative answers can be found from:

❯ nslookup mta5.am0.yahoodns.net 8.8.8.8

Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.76
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.75
Name:   mta5.am0.yahoodns.net
Address: 67.195.204.79
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.94
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.77
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.111
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.110
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.106
```

3. Tracing DNS with Wireshark

Part 3A:

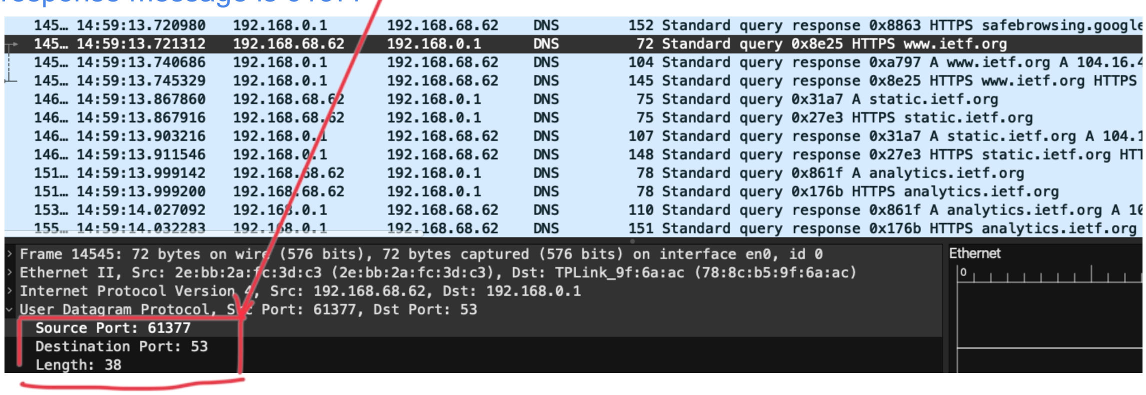
4. Locate the DNS query and response messages. Are then sent over UDP or TCP?

They are sent over UDP



5. What is the destination port for the DNS query message? What is the source port of DNS response message?

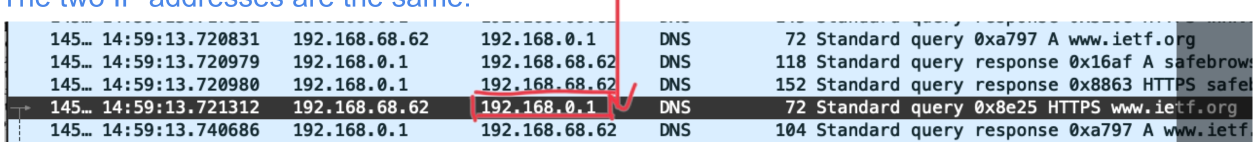
The destination port for a DNS query message is 53, and the source port of the DNS response message is 61377



6. To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?

Query message sent to IP address: 192.168.0.1.

The two IP addresses are the same.



```
❯ scutil --dns
DNS configuration

resolver #1
  nameserver[0] : 192.168.0.1
  nameserver[1] : 192.168.68.1
  if_index : 11 (en0)
  flags    : Request A records
  reach    : 0x00000002 (Reachable)
```

7. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

It is a Type A query; the Queries message does not contain any answers.

```
⌄ Queries
  ⌄ www.ietf.org: type A, class IN
      Name: www.ietf.org
      [Name Length: 12]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
```

8. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

2 answers are provided, and each answers contain Name, Type, Class, Time to live, Data length, and Address.

```
Answers
1  www.ietf.org: type A, class IN, addr 104.16.44.99
       Name: www.ietf.org
       Type: A (1) (Host Address)
       Class: IN (0x0001)
       Time to live: 242 (4 minutes, 2 seconds)
       Data length: 4
       Address: 104.16.44.99
2  www.ietf.org: type A, class IN, addr 104.16.45.99
       Name: www.ietf.org
       Type: A (1) (Host Address)
       Class: IN (0x0001)
       Time to live: 242 (4 minutes, 2 seconds)
       Data length: 4
       Address: 104.16.45.99
   [Request In: 14542]
   [Time: 0.019855000 seconds]
```

9. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?
The first SYN packet was sent to 104.16.45.99, which corresponds to the first IP address provided in the DNS response messages.

10. This web page contains images. Before retrieving each image, does your host issue new DNS queries? No

Part 3B: Do an *nslookup* on www.mit.edu

11. What is the destination port for the DNS query message? What is the source port of DNS response message?
Destination port: 53; Source port: 49236

12. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?
It was sent to the IP address 192.168.0.1; we can see from the screenshot below it is the default local DNS server.

ip.addr == 192.168.68.62

No.	Time	Source	Destination	Protocol	Length	Info
209	15:33:25.938137	192.168.68.62	192.168.0.1	DNS	71	Standard query 0xb15b A www.mit.edu
210	15:33:25.966240	192.168.0.1	192.168.68.62	DNS	160	Standard query response 0xb15b A www.mit.e

```
❯ scutil --dns
DNS configuration

resolver #1
  nameserver[0] : 192.168.0.1
  nameserver[1] : 192.168.68.1
  if_index : 11 (en0)
  flags    : Request A records
  reach    : 0x00000002 (Reachable)
```

13. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

It is Type A, the query message doesn't contain any answers.

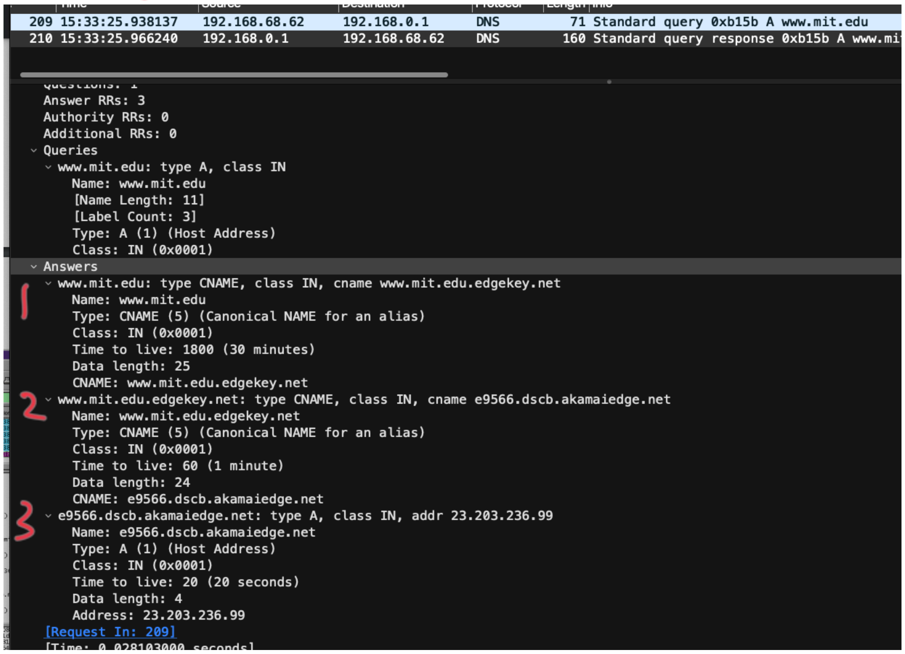
ip.addr == 192.168.68.62

No.	Time	Source	Destination	Protocol	Length	Info
209	15:33:25.938137	192.168.68.62	192.168.0.1	DNS	71	Standard query 0xb15b A www.mit.edu
210	15:33:25.966240	192.168.0.1	192.168.68.62	DNS	160	Standard query response 0xb15b A www.m

```
    [Checksum Status: Unverified]
    [Stream index: 4]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (29 bytes)
v Domain Name System (query)
    Transaction ID: 0xb15b
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    v www.mit.edu: type A, class IN
        Name: www.mit.edu
        [Name Length: 11]
        [Label Count: 3]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
    [Response In: 210]
```

14. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

3 answers are provided, each answer contains: Name, Type, Class, Time to live, Data length, CNAME, and address.

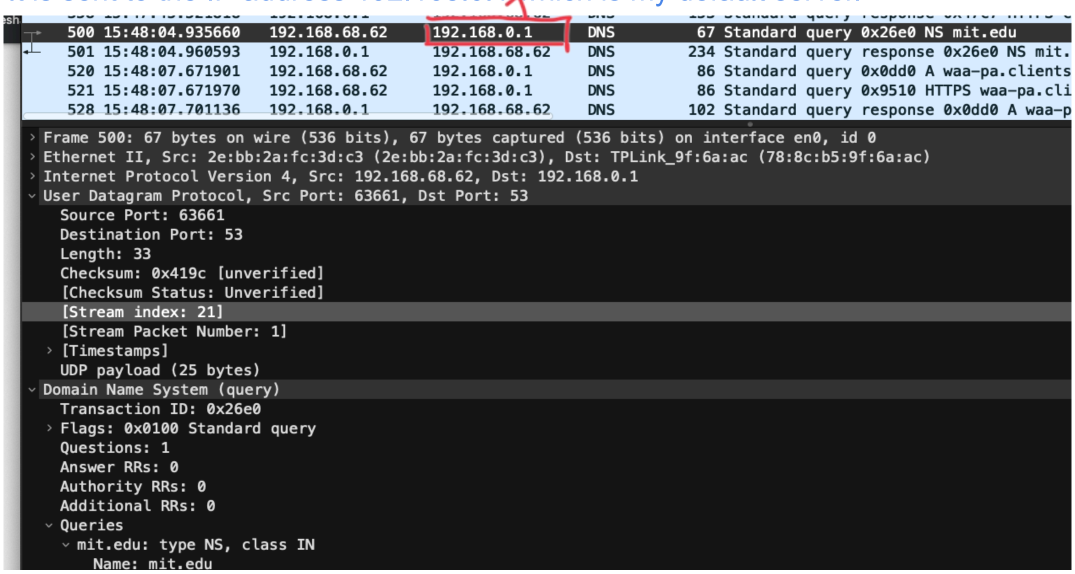


15. Provide a screenshot.

Part 3C: Do an nslookup –type=NS mit.edu

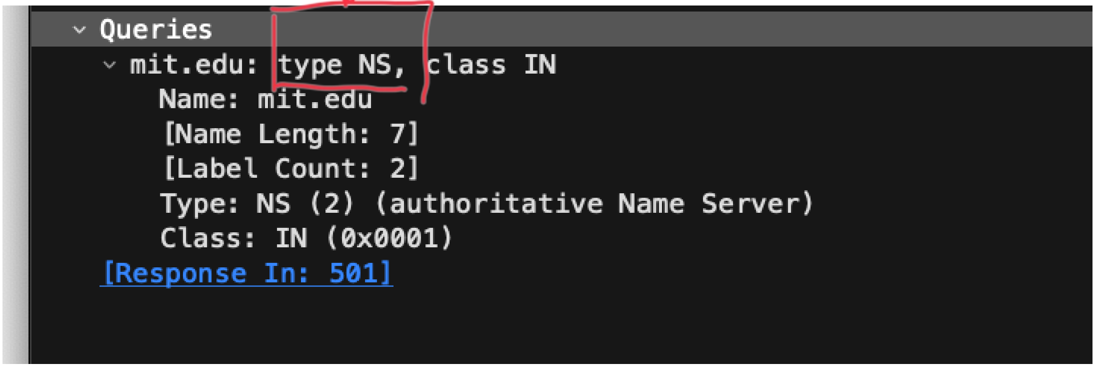
16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

It is sent to the IP address 192.168.0.1 which is my default server.



17. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

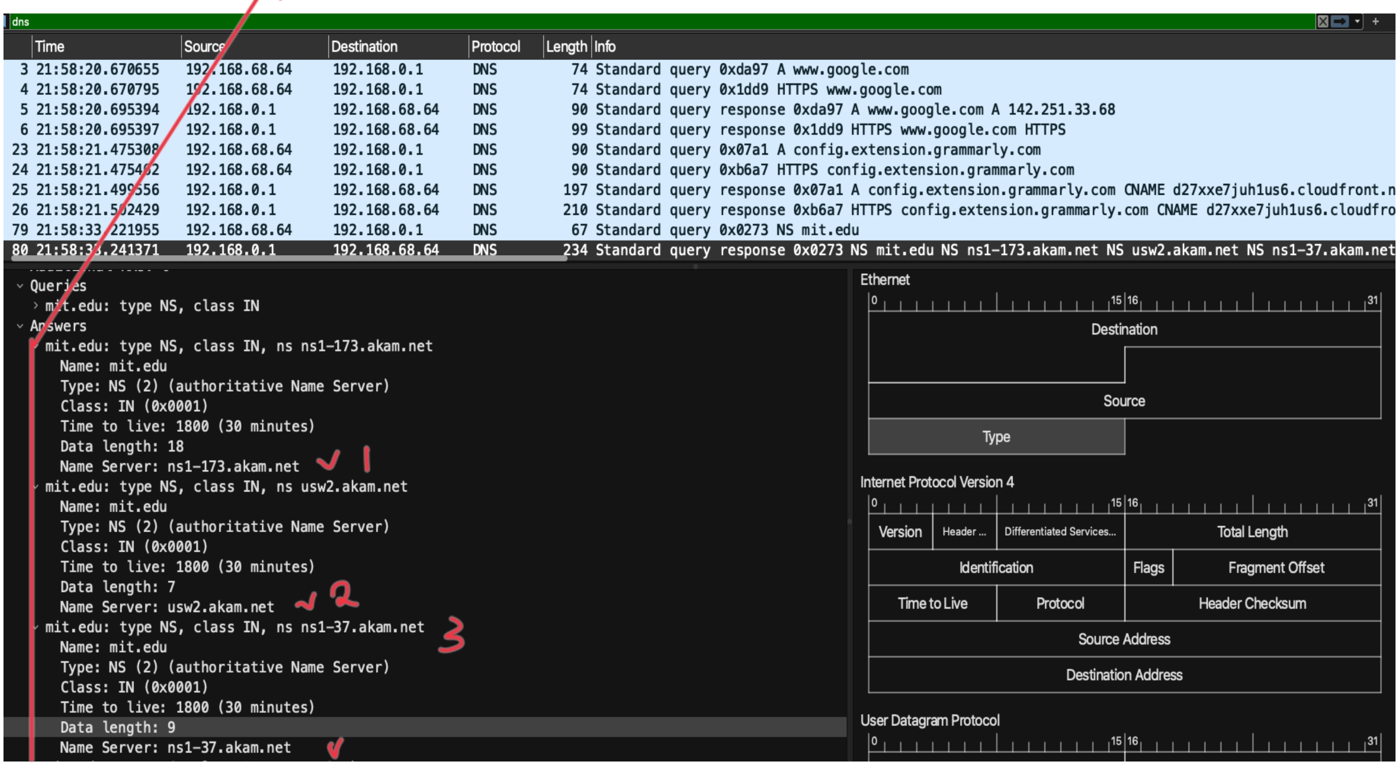
Type: NS, the query doesn't contain any answers

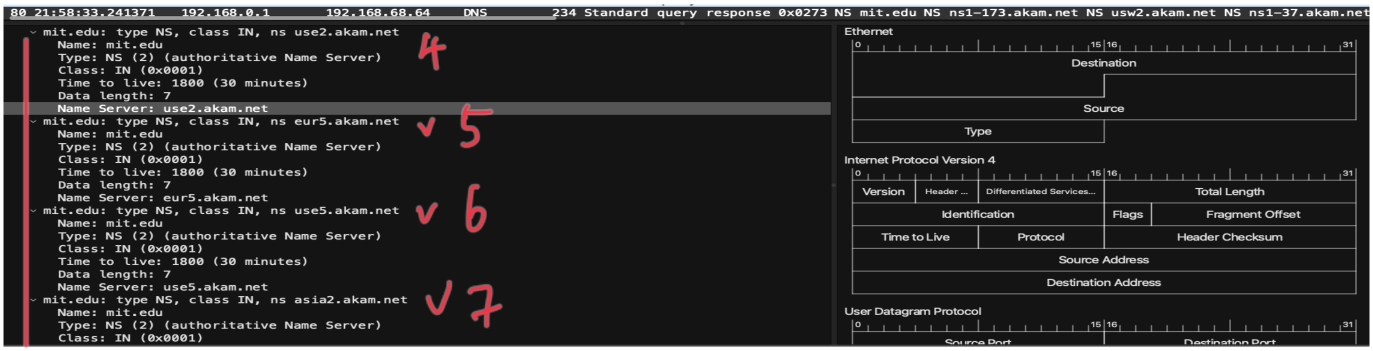


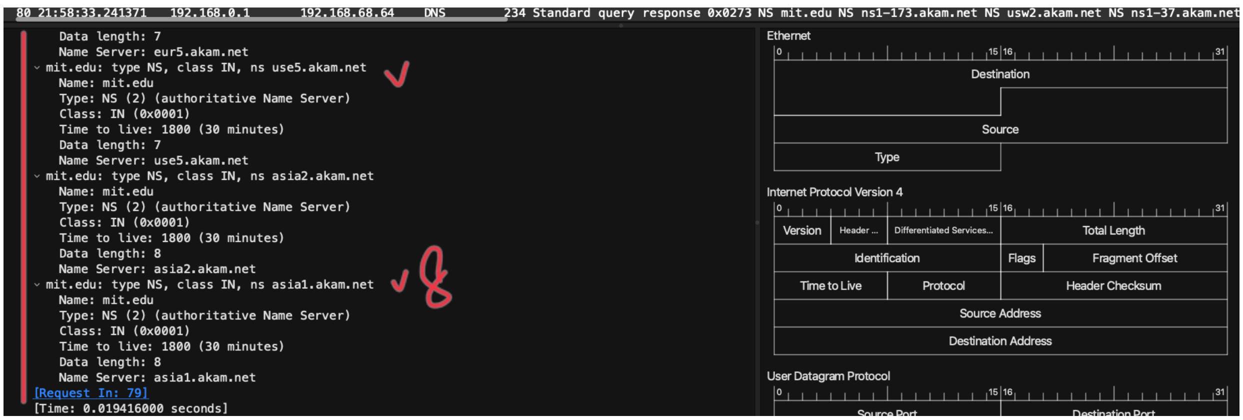
18. Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT namesers?

It provided 8 nameservers:

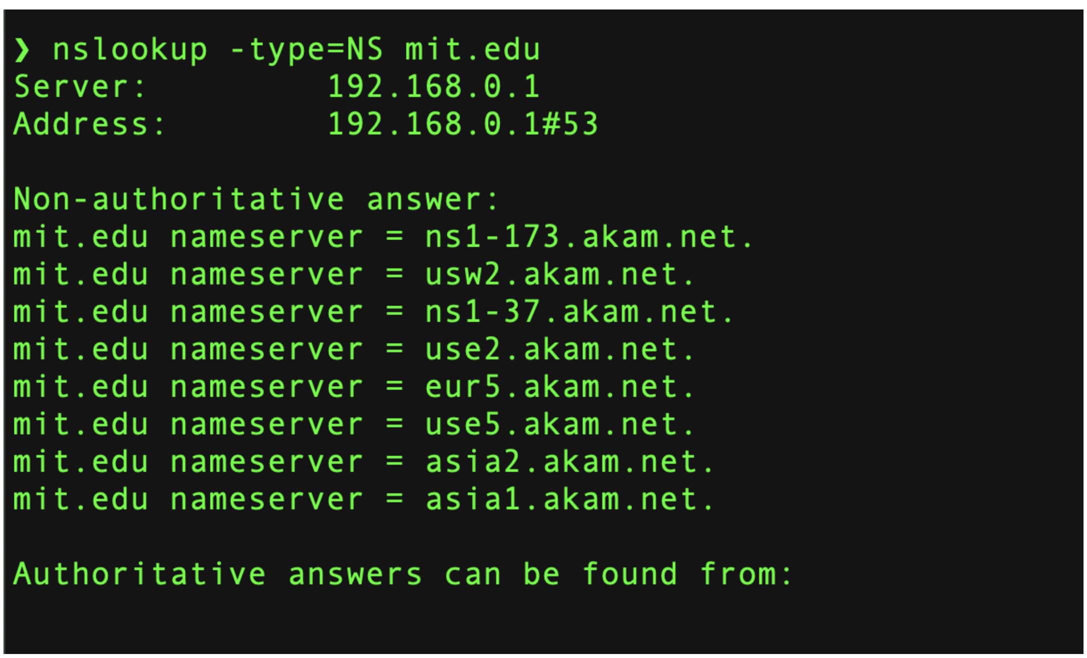
mit.edu	nameserver = ns1-173.akam.net. 1
mit.edu	nameserver = usw2.akam.net. 2
mit.edu	nameserver = ns1-37.akam.net. 3
mit.edu	nameserver = use2.akam.net. 4
mit.edu	nameserver = eur5.akam.net. 5
mit.edu	nameserver = use5.akam.net. 6
mit.edu	nameserver = asia2.akam.net. 7
mit.edu	nameserver = asia1.akam.net. 8







19. Provide a screenshot.



Part 3D: nslookup www.aiit.or.kr bitsy.mit.edu

20. To what IP address is the DNS query message sent? Is this the IP address of your

default local DNS server? If not, what does the IP address correspond to?

This DNS query message is sent to 192.168.01 which is the IP address of the MIT DNS response sender

No.	Time	Source	Destination	Protocol	Length	Info
27	21:30:06.682235	192.168.68.64	192.168.0.1	DNS	73	Standard query 0x3e6e A bitsy.mit.edu
28	21:30:06.704558	192.168.0.1	192.168.68.64	DNS	89	Standard query response 0x3e6e A bitsy.mit.edu A 18.0.72.3
31	21:30:06.708696	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr
92	21:30:11.709540	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr

21. Examine the DNS query message. What “Type” of DNS query is it? Does the Does query message contains any “answers”?

Type: Type A, the query message doesn’t contain any answers

```
Additional RRs: 0
⌄ Queries
  ⌄ bitsy.mit.edu: type A, class IN
      Name: bitsy.mit.edu
      [Name Length: 13]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
```

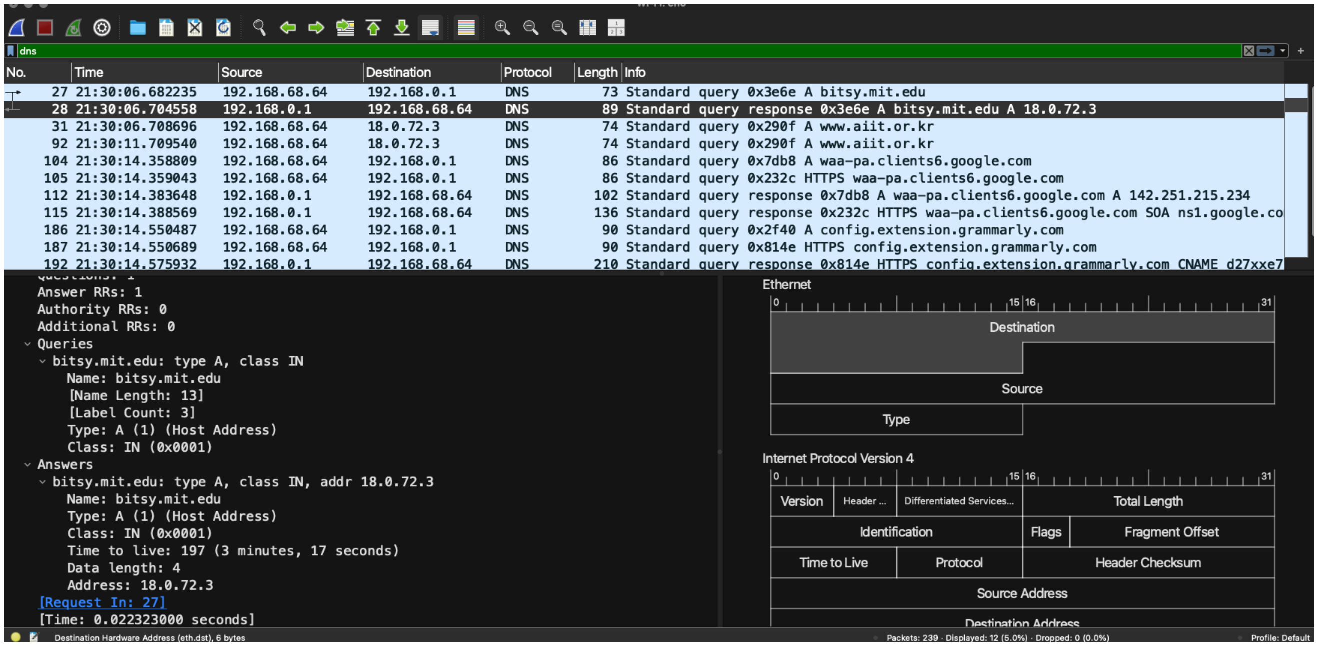
22. Examine the DNS response message. How many “answers” are provided? What does each of these answers contain?

One answer is provided, and the answer contains Name, type, Class, Time to live, Data length, and Address.

```
⌄ Answers
  ⌄ bitsy.mit.edu: type A, class IN, addr 18.0.72.3
      Name: bitsy.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 197 (3 minutes, 17 seconds)
      Data length: 4
      Address: 18.0.72.3
  [Request In: 27]
  [Time: 0.022323000 seconds]
```

1

23. Provide a screenshot.


dns
No. Time Source Destination Protocol Length Info
27 21:30:06.682235 192.168.68.64 192.168.0.1 DNS 73 Standard query 0x3e6e A bitsy.mit.edu
28 21:30:06.704558 192.168.0.1 192.168.68.64 DNS 89 Standard query response 0x3e6e A bitsy.mit.edu A 18.0.72.3
31 21:30:06.708696 192.168.68.64 18.0.72.3 DNS 74 Standard query 0x290f A www.aiit.or.kr
92 21:30:11.709540 192.168.68.64 18.0.72.3 DNS 74 Standard query 0x290f A www.aiit.or.kr
104 21:30:14.358809 192.168.68.64 192.168.0.1 DNS 86 Standard query 0x7db8 A waa-pa.clients6.google.com
105 21:30:14.359043 192.168.68.64 192.168.0.1 DNS 86 Standard query 0x232c HTTPS waa-pa.clients6.google.com
112 21:30:14.383648 192.168.0.1 192.168.68.64 DNS 102 Standard query response 0x7db8 A waa-pa.clients6.google.com A 142.251.215.234
115 21:30:14.388569 192.168.0.1 192.168.68.64 DNS 136 Standard query response 0x232c HTTPS waa-pa.clients6.google.com SOA ns1.google.co
186 21:30:14.550487 192.168.68.64 192.168.0.1 DNS 90 Standard query 0x2f40 A config.extension.grammarly.com
187 21:30:14.550689 192.168.68.64 192.168.0.1 DNS 90 Standard query 0x814e HTTPS config.extension.grammarly.com
192 21:30:14.575932 192.168.0.1 192.168.68.64 DNS 210 Standard query response 0x814e HTTPS config.extension.grammarly.com CNAME d27xxe7
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries
bitsy.mit.edu: type A, class IN
Name: bitsy.mit.edu
[Name Length: 13]
[Label Count: 3]
Type: A (1) (Host Address)
Class: IN (0x0001)
Answers
bitsy.mit.edu: type A, class IN, addr 18.0.72.3
Name: bitsy.mit.edu
Type: A (1) (Host Address)
Class: IN (0x0001)
Time to live: 197 (3 minutes, 17 seconds)
Data length: 4
Address: 18.0.72.3
[Request In: 27]
[Time: 0.022323000 seconds]
Ethernet
Destination
Source
Type
Internet Protocol Version 4
Version
Header ...
Differentiated Services...
Total Length
Identification
Flags
Fragment Offset
Time to Live
Protocol
Header Checksum
Source Address
Destination Address
Destination Hardware Address (eth.dst), 6 bytes
Packets: 239 · Displayed: 12 (5.0%) · Dropped: 0 (0.0%)
Profile: Default